

[HCI/2024/Q2]

2 Name your Jupyter Notebook as

TASK2_<your name>_<centre number>_<index number>.ipynb

For this question, you must **not** use any built-in search or sort function.

When RGB values are given out of 255, each number represents the intensity of the respective colour channel (Red, Green, or Blue) on a scale from 0 to 255.

When combined, these three values (R, G, B) define a specific colour in the RGB colour space. For example, RGB (255, 0, 0) represents pure red, RGB (0, 255, 0) represents pure green, and RGB (0, 0, 255) represents pure blue. Different combinations of these values create the wide spectrum of colours.

A paint company has a catalogue of 15 colours and their RGB values are stored in a csv file `paint.csv`. For example, 3, 126, 251, ocean means that the colour named ocean is RGB (3, 126, 251).

For each of the sub-tasks, add a comment statement at the beginning of the code using the hash symbol '#', to indicate the sub-task the program code belongs to, for example:

In [1]:

```
#Task 2.1  
Program Code
```

Output:

Task 2.1

Write a function `read_csv(filename)` which

- reads a csv file, `filename`
- stores the records in a 2D-array with 4 columns `red`, `green`, `blue` and `name`
- stores `red`, `green` and `blue` as integers
- stores `name` as a string
- returns the array.

Use the following code to test your function.

```
colours = read_csv("paint.csv")  
print(len(colours))  
print(colours)
```

[4]

[HCI/2024/Q2 Cont']

Task 2.2

Write a function `find(array, r, g, b)` which

- linearly searches `array` for a colour with `r, g, b` values
- returns name or `'nil'` if the colour is not found.

Use the following code to test your function.

```
print(find(colours, 253, 253, 5))  
print(find(colours, 5, 253, 253))
```

 [4]

Task 2.3

When a colour has a larger value for their Red colour channel, we say that the colour is more red. Similarly, colours can be more blue and more green.

Write a function `rgb(array, colour)` to implement a bubble sort algorithm which

- sorts `array` in descending order from the highest to lowest intensity by the `colour` provided
- takes `colour` as one of the values `'red', 'green' or 'blue'`
- prints an appropriate error message and returns an empty array if an invalid `colour` is given
- returns the sorted array.

Use the following code to test your function.

```
print(rgb(colours, 'yellow'))  
print(rgb(colours, 'blue'))
```

 [6]

Task 2.4

As the colour intensity increases, the brightness also increases, we say that a colour is brighter when the total of their three colour channels gives a larger value.

Write a function `bright(array)` to implement a quick sort algorithm which

- sorts `array` in descending order from the brightest colour to the darkest colour
- returns the sorted array.

Use the following code to test your function.

```
print(bright(colours))
```

 [6]

[HCI/2024/Q2 Cont']

Task 2.5

The paint company wants to create a menu which allows their clients to choose their preferred colour and recommends paint colours based on their preference.

Write a function `menu()` which

- reads the file 'paint.csv'
- allows the user to choose their preference: red, green, blue, dark or bright
- repeats the prompt until a valid input is received
- prints out the name of the top 3 colours which matches the preference.

Use your function to show the recommended colours for **three** different users who prefer red, bright and dark colours respectively. [6]

Save your Jupyter Notebook for Task 2.